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Sem: III AY: 2026-27

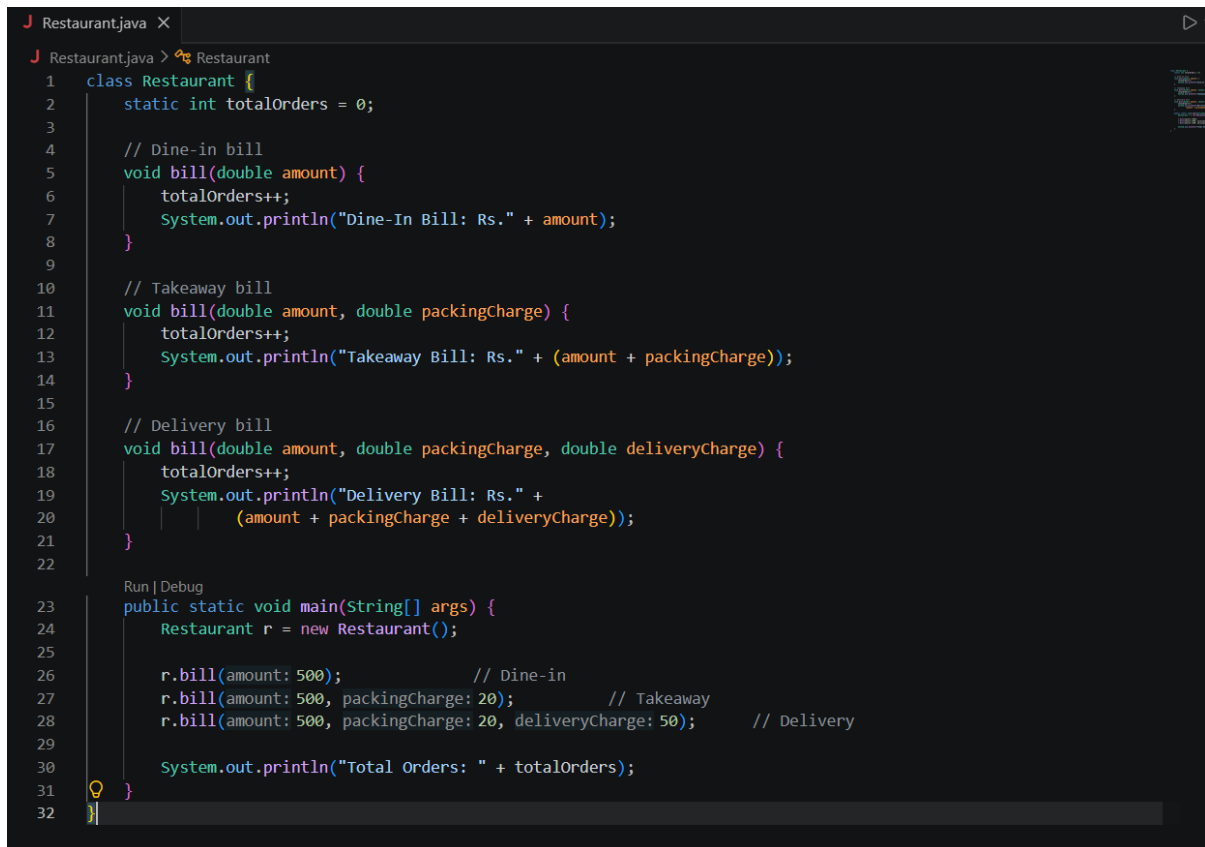
TITLE: Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

- ```
Calculator.java X
Calculator.java > Calculator
1 class Calculator {
2 static int count = 0;
3
4 // Addition of integers
5 int add(int a, int b) {
6 count++;
7 return a + b;
8 }
9
10 // Addition of decimals
11 double add(double a, double b) {
12 count++;
13 return a + b;
14 }
15
16 Run | Debug
17 public static void main(String[] args) {
18 Calculator c = new Calculator();
19
20 System.out.println("Integer Addition: " + c.add(a: 10, b: 20));
21 System.out.println("Decimal Addition: " + c.add(a: 10.5, b: 20.5));
22
23 System.out.println("Total Calculations: " + count);
24 }
```

OUTPUT:

```
[Running] cd "c:\Users\Arpit\java_assignments\" && javac Calculator.java && java Calculator
Integer Addition: 30
Decimal Addition: 31.0
Total Calculations: 2
```

2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.



```
J Restaurant.java X
J Restaurant.java > Restaurant
1 class Restaurant {
2 static int totalOrders = 0;
3
4 // Dine-in bill
5 void bill(double amount) {
6 totalOrders++;
7 System.out.println("Dine-In Bill: Rs." + amount);
8 }
9
10 // Takeaway bill
11 void bill(double amount, double packingCharge) {
12 totalOrders++;
13 System.out.println("Takeaway Bill: Rs." + (amount + packingCharge));
14 }
15
16 // Delivery bill
17 void bill(double amount, double packingCharge, double deliveryCharge) {
18 totalOrders++;
19 System.out.println("Delivery Bill: Rs." +
20 (amount + packingCharge + deliveryCharge));
21 }
22
23 Run | Debug
24 public static void main(String[] args) {
25 Restaurant r = new Restaurant();
26
27 r.bill(amount: 500); // Dine-in
28 r.bill(amount: 500, packingCharge: 20); // Takeaway
29 r.bill(amount: 500, packingCharge: 20, deliveryCharge: 50); // Delivery
30
31 System.out.println("Total Orders: " + totalOrders);
32 }
33 }
```

OUTPUT:

```
[Running] cd "c:\Users\Arpit\java_assignments\" && javac Restaurant.java && java Restaurant
Dine-In Bill: Rs.500.0
Takeaway Bill: Rs.520.0
Delivery Bill: Rs.570.0
Total Orders: 3
```